## Higher National Diploma in Engineering (Civil)

**4th Year, First Semester Examination – 2018**

## CE 3224 - Environmental Engineering

| | |
|---|---|
| Instructions for Candidates: | No. of questions : 06 |
| **Answer five questions only** | No. of pages : 02 |
| **All questions carry equal marks** | Time :3 hrs |

Q1:

a. Write down four (4) objectives of wastewater treatment. (4 Marks)

b. Name the main three types of organic compounds present in the wastewater? (3 Marks)

c. State the three (03) uses of Biological Oxygen Demand (BOD) in wastewater treatment. (3 Marks)

d. Briefly describe about three weaknesses in the current wastewater treatment techniques. (6 Marks)

e. "Alkalinity in wastewater is important for the treatment processes of wastewater". Explain this statement. (4 Marks)

(Total 20 Marks)

Q2.

a. Write down three (03) odour controlling methods used in wastewater treatment. (3 Marks)

b. What is the purpose of secondary treatment process and briefly explain the main three (03) biological processes that are used in this treatment. (7 Marks)

c. With a neat sketch describe the treatment process happening in the trickling filter. (6 Marks)

d. Write down four objectives of sludge stabilization. (4 Marks)

(Total 20 Marks)

Q3.

a. Briefly describe three mechanisms of materials deterioration attributed by air pollution. (6 Marks)

b. "Air pollution has no boundaries". Explain this statement. (3 Marks)

c. Write down five (05) compounds and their sources that cause for the indoor air pollution. (5 Marks)

d. Write down three equipments that can be used for the removal of the particulate pollutants from air. (3 Marks)

e. What is "Kyoto Protocol"? (3 Marks)

(Total 20 Marks)

Q4.

a. Mention five broad categories of Municipal Solid Waste (MSW) and give two examples for each. (5 Marks)

b. What do you mean by the sanitary land filling? (3 Marks)

c. What are the environmental considerations that have to be considered when designing the landfills? (3 Marks)

d. Propose a proper solid waste management system for the hospital. (9 Marks)

(Total 20 Marks)

Q5.

a. What is mean by the Hearing Threshold Level (HTL)? (3 Marks)

b. What are the main three types of noise? (3 Marks)

c. Write down four (04) factors that should be considered prior to control the noise. (8 Marks)

d. Write down noise control methods for following noises at the source. (6 marks)

I. Structure borne noise

II. Air borne noise

III. Reverberation (Total 20 Marks)

Q6.

a. What is meant by the sustainable development? (2 Marks)

b. Briefly describe how unsustainable buildings are adversely impact on biodiversity. (4 Marks)

c. In Green building concept, write down five (05) methods that can be used to increase the indoor environmental quality. (5 Marks)

d. What is the difference between Initial Environmental Examination (IEE) and Environmental Impact Assessment (EIA)? (3 Marks)

e. Write down the main steps of EIA. (6 Marks)

(Total 20 Marks)